POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6013D | Semester VI | Theory | 4 Credits

Habitat Technology

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6013D as Habitat Technology in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Roorkee’s Sustainable Architecture course, the principles of the Laurie Baker Center (LBC) and polytechnic cost-effective construction curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Building designs, material selections, structural modifications and housing policy applications must be based on approved engineering designs, certified material testing, current building codes and competent professional review.

Verified source note: Verified sources support sustainable architecture, climatic design, vernacular architecture, cost-effective materials (mud, bamboo, laterite), Laurie Baker's techniques (rat-trap bond, filler slabs) and affordable housing principles. The four-module sequence is a learning sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Habitat Technology studies the design and construction of human settlements using cost-effective, sustainable and climate-responsive techniques. It develops the ability to utilize local materials (mud, bamboo, laterite), apply Laurie Baker's innovative building methods, plan climate-appropriate structures, and understand affordable housing policies while respecting the technical and cultural boundaries of habitat development.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Building construction, construction materials, environmental studies, basic surveying and architectural awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain human settlement fundamentals, housing needs and the principles of sustainable habitat development.
2. Explain cost-effective building materials, including local resources like mud, bamboo, laterite and industrial by-products.
3. Apply Laurie Baker's innovative techniques, including rat-trap bond, filler slabs, and cost-effective arches and domes.
4. Explain climate-responsive design, energy efficiency in buildings, and rural/urban housing policies and schemes.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO ▯▯▯▯▯▯ exam-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. Define, explain, apply ▯▯▯▯▯ action words ▯▯▯▯▯▯▯▯▯▯▯▯.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain human settlement patterns, housing needs and the concept of sustainable habitats.	Understand
CO2	Explain the properties and applications of cost-effective and local building materials.	Understand
CO3	Apply Laurie Baker's cost-effective construction techniques for sustainable building.	Apply
CO4	Explain climate-responsive design principles and the framework of affordable housing policies.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Human Settlements and Housing Needs	Evolution of human settlements; urbanization and its impacts; housing as a basic need; assessment of housing shortage; principles of sustainable habitat development; UN Sustainable Development Goals (SDG 11).	Approx. 14 hours	CO1	Module 1
2	Cost-Effective and Local Building Materials	Mud as a building material (stabilized soil blocks, cob, adobe); bamboo and its structural applications; laterite and local stone; industrial by-products (fly ash, GGBS); recycled aggregates and C&D waste.	Approx. 16 hours	CO2	Module 2
3	Laurie Baker's Innovative Techniques	Laurie Baker's philosophy (simplicity, localism, economy); Rat-trap bond masonry; filler slabs (types of fillers); cost-effective arches, domes and vaults; brick jalis and ornamentation.	Approx. 16 hours	CO2	Module 3
4	Climate-Responsive Design and Housing Policy	Climatic zones and building design; thermal comfort; passive heating and cooling; energy efficiency in buildings; rural and urban housing policies; affordable housing schemes (PMAY).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Human Settlements and Housing Needs

Evolution of human settlements; urbanization and its impacts; housing as a basic need; assessment of housing shortage; principles of sustainable habitat development; UN Sustainable Development Goals (SDG 11).

CO1 · Approx. 14 hours

Cost-Effective and Local Building Materials

Mud as a building material (stabilized soil blocks, cob, adobe); bamboo and its structural applications; laterite and local stone; industrial by-products (fly ash, GGBS); recycled aggregates and C&D waste.

CO2 · Approx. 16 hours

Laurie Baker's Innovative Techniques

Laurie Baker's philosophy (simplicity, localism, economy); Rat-trap bond masonry; filler slabs (types of fillers); cost-effective arches, domes and vaults; brick jalis and ornamentation.

CO2 · Approx. 16 hours

Climate-Responsive Design and Housing Policy

Climatic zones and building design; thermal comfort; passive heating and cooling; energy efficiency in buildings; rural and urban housing policies; affordable housing schemes (PMAY).

CO4 · Approx. 14 hours

MODULE 1

Human Settlements and Housing Needs

Evolution of human settlements; urbanization and its impacts; housing as a basic need; assessment of housing shortage; principles of sustainable habitat development; UN Sustainable Development Goals (SDG 11).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

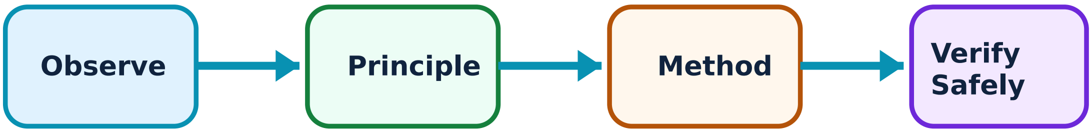
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Human Settlements and Housing Needs: identify the system, state its principle, follow the correct method and verify the result safely.

1. Settlements and urbanization−

Human settlements evolve from rural to urban forms, influenced by economic, social and environmental factors. Rapid urbanization leads to housing shortages, slums and environmental stress. Sustainable development aims to make cities and settlements inclusive, safe, resilient and sustainable.

Study and application method

List the major characteristics of rural and urban settlements; identify three impacts of rapid urbanization on the housing sector.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the major characteristics of rural and urban settlements; identify three impacts of rapid urbanization on the housing sector.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Urbanization patterns are complex. Do not assume generic housing solutions will work in all cultural or geographic contexts without authorised local studies.

2. Sustainable habitat principles−

Sustainable habitats balance human needs with environmental protection. Principles include land-use efficiency, water conservation, waste management, use of local materials and social inclusivity. SDG 11 provides the global framework for sustainable cities and communities.

Study and application method

Describe four principles of a sustainable habitat; explain the importance of community participation in habitat planning.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe four principles of a sustainable habitat; explain the importance of community participation in habitat planning.

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Common mistake / precaution: Sustainability requires a holistic approach. Do not prioritise one aspect (e.g., energy) at the expense of others (e.g., social equity or structural safety).

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Cost-Effective and Local Building Materials

Mud as a building material (stabilized soil blocks, cob, adobe); bamboo and its structural applications; laterite and local stone; industrial by-products (fly ash, GGBS); recycled aggregates and C&D waste.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

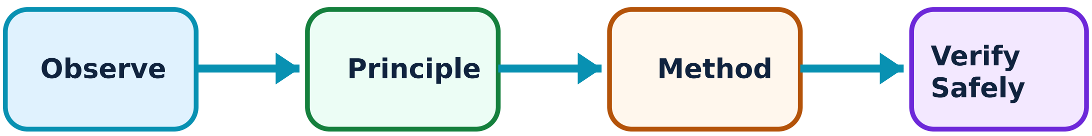
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Cost-Effective and Local Building Materials: identify the system, state its principle, follow the correct method and verify the result safely.

1. Earth and bamboo construction−

Earth (mud) is a traditional, low-embodied-energy material. Stabilized Earth Blocks (SEB) improve durability. Bamboo is a fast-growing, high-tensile-strength renewable resource suitable for structural elements, scaffolding and reinforcement.

Study and application method

Compare the properties of adobe and stabilized soil blocks; sketch three structural joints used in bamboo construction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the properties of adobe and stabilized soil blocks; sketch three structural joints used in bamboo construction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Earth and bamboo are sensitive to moisture and biological attack. All materials must be properly treated and protected as per authorised standards.

2. Local and alternative materials−

Laterite and local stones reduce transport costs. Industrial by-products like fly ash and GGBS can replace cement in concrete and bricks. Using Construction and Demolition (C&D) waste as recycled aggregate conserves natural resources.

Study and application method

List the advantages of using fly ash bricks over conventional clay bricks; identify two local building materials available in your region.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the advantages of using fly ash bricks over conventional clay bricks; identify two local building materials available in your region.

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Common mistake / precaution: Alternative materials must meet structural and durability requirements. Do not use unverified waste materials for safety-critical structural components.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Laurie Baker's Innovative Techniques

Laurie Baker's philosophy (simplicity, localism, economy); Rat-trap bond masonry; filler slabs (types of fillers); cost-effective arches, domes and vaults; brick jalis and ornamentation.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

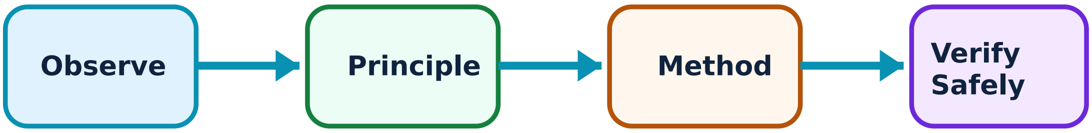
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Laurie Baker's Innovative Techniques: identify the system, state its principle, follow the correct method and verify the result safely.

1. Rat-trap bond and filler slabs−

Rat-trap bond is a brick-laying method that creates a cavity, saving bricks and mortar while providing thermal insulation. Filler slabs replace part of the concrete in the tension zone with low-cost materials (tiles, pots), reducing weight and cost without compromising strength.

Study and application method

Sketch the plan and elevation of a wall in rat-trap bond; explain how a filler slab reduces the cost and dead weight of a roof.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the plan and elevation of a wall in rat-trap bond; explain how a filler slab reduces the cost and dead weight of a roof.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Innovative techniques require skilled workmanship. Do not implement these methods without training and supervision by authorised personnel.

2. Arches, domes and jalis−

Arches and domes use the compressive strength of bricks to span openings and spaces cost-effectively. Brick jalis (perforated walls) provide natural light and ventilation (the 'venturi effect') while maintaining privacy and reducing material use.

Study and application method

Draw a conceptual sketch of a brick jali pattern and explain its role in natural ventilation; describe the benefits of using a brick arch over a concrete lintel.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a conceptual sketch of a brick jali pattern and explain its role in natural ventilation; describe the benefits of using a brick arch over a concrete lintel.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Arches and domes involve complex structural forces. Design and centering must be performed by competent personnel to prevent structural failure.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Climate-Responsive Design and Housing Policy

Climatic zones and building design; thermal comfort; passive heating and cooling; energy efficiency in buildings; rural and urban housing policies; affordable housing schemes (PMAY).

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

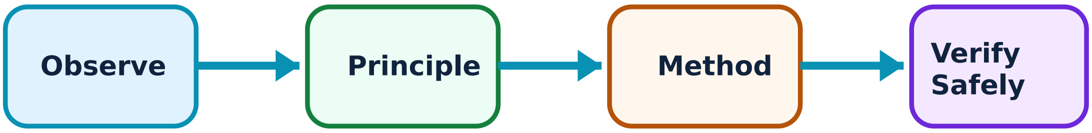
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Climate-Responsive Design and Housing Policy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Climate-responsive architecture−

Design must respond to local climate (hot-dry, warm-humid, etc.) to ensure thermal comfort with minimal energy. Techniques include orientation, shading, natural ventilation, thermal mass and insulation. Vernacular architecture provides many time-tested examples.

Study and application method

Identify the key design strategies for a building in a warm-humid climate; explain the concept of 'thermal mass' in climate control.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify the key design strategies for a building in a warm-humid climate; explain the concept of 'thermal mass' in climate control.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Climate-responsive design must not compromise indoor air quality or structural safety. Use only authorised climatic data for design calculations.

2. Housing policies and schemes−

Government policies and schemes (like Pradhan Mantri Awas Yojana - PMAY) aim to provide affordable housing for all. These include financial assistance, slum redevelopment and public-private partnerships to bridge the housing gap in rural and urban areas.

Study and application method

List the major components of the PMAY housing scheme; describe the role of government in promoting cost-effective habitat technology.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the major components of the PMAY housing scheme; describe the role of government in promoting cost-effective habitat technology.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Policy implementation involves legal and financial regulations. Do not treat a study-level summary as an authorised guide for housing finance or legal eligibility.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Human Settlements and Housing Needs

Use the concept flow to revise observation, principle, method and verification.

Module 2: Cost-Effective and Local Building Materials

Use the concept flow to revise observation, principle, method and verification.

Module 3: Laurie Baker's Innovative Techniques

Use the concept flow to revise observation, principle, method and verification.

Module 4: Climate-Responsive Design and Housing Policy

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare rat-trap bond and English bond in a conceptual table showing material savings.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a cross-section of a filler slab identifying the filler materials and reinforcement.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify five cost-effective building techniques used in a vernacular building in your region.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a layout for a climate-responsive house for a warm-humid region.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a summary of a local or national affordable housing scheme.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Human Settlements and Housing Needs

Explain Settlements and urbanization and state one correct method, application or precaution.−

Human settlements evolve from rural to urban forms, influenced by economic, social and environmental factors. Rapid urbanization leads to housing shortages, slums and environmental stress. Sustainable development aims to make cities and settlements inclusive, safe, resilient and sustainable.

Answer framework: List the major characteristics of rural and urban settlements; identify three impacts of rapid urbanization on the housing sector.

Important point: Urbanization patterns are complex. Do not assume generic housing solutions will work in all cultural or geographic contexts without authorised local studies.

Explain Sustainable habitat principles and state one correct method, application or precaution.−

Sustainable habitats balance human needs with environmental protection. Principles include land-use efficiency, water conservation, waste management, use of local materials and social inclusivity. SDG 11 provides the global framework for sustainable cities and communities.

Answer framework: Describe four principles of a sustainable habitat; explain the importance of community participation in habitat planning.

Important point: Sustainability requires a holistic approach. Do not prioritise one aspect (e.g., energy) at the expense of others (e.g., social equity or structural safety).

Module 2 — Cost-Effective and Local Building Materials

Explain Earth and bamboo construction and state one correct method, application or precaution.−

Earth (mud) is a traditional, low-embodied-energy material. Stabilized Earth Blocks (SEB) improve durability. Bamboo is a fast-growing, high-tensile-strength renewable resource suitable for structural elements, scaffolding and reinforcement.

Answer framework: Compare the properties of adobe and stabilized soil blocks; sketch three structural joints used in bamboo construction.

Important point: Earth and bamboo are sensitive to moisture and biological attack. All materials must be properly treated and protected as per authorised standards.

Explain Local and alternative materials and state one correct method, application or precaution.−

Laterite and local stones reduce transport costs. Industrial by-products like fly ash and GGBS can replace cement in concrete and bricks. Using Construction and Demolition (C&D) waste as recycled aggregate conserves natural resources.

Answer framework: List the advantages of using fly ash bricks over conventional clay bricks; identify two local building materials available in your region.

Important point: Alternative materials must meet structural and durability requirements. Do not use unverified waste materials for safety-critical structural components.

Module 3 — Laurie Baker's Innovative Techniques

Explain Rat-trap bond and filler slabs and state one correct method, application or precaution.−

Rat-trap bond is a brick-laying method that creates a cavity, saving bricks and mortar while providing thermal insulation. Filler slabs replace part of the concrete in the tension zone with low-cost materials (tiles, pots), reducing weight and cost without compromising strength.

Answer framework: Sketch the plan and elevation of a wall in rat-trap bond; explain how a filler slab reduces the cost and dead weight of a roof.

Important point: Innovative techniques require skilled workmanship. Do not implement these methods without training and supervision by authorised personnel.

Explain Arches, domes and jalis and state one correct method, application or precaution.−

Arches and domes use the compressive strength of bricks to span openings and spaces cost-effectively. Brick jalis (perforated walls) provide natural light and ventilation (the 'venturi effect') while maintaining privacy and reducing material use.

Answer framework: Draw a conceptual sketch of a brick jali pattern and explain its role in natural ventilation; describe the benefits of using a brick arch over a concrete lintel.

Important point: Arches and domes involve complex structural forces. Design and centering must be performed by competent personnel to prevent structural failure.

Module 4 — Climate-Responsive Design and Housing Policy

Explain Climate-responsive architecture and state one correct method, application or precaution.−

Design must respond to local climate (hot-dry, warm-humid, etc.) to ensure thermal comfort with minimal energy. Techniques include orientation, shading, natural ventilation, thermal mass and insulation. Vernacular architecture provides many time-tested examples.

Answer framework: Identify the key design strategies for a building in a warm-humid climate; explain the concept of 'thermal mass' in climate control.

Important point: Climate-responsive design must not compromise indoor air quality or structural safety. Use only authorised climatic data for design calculations.

Explain Housing policies and schemes and state one correct method, application or precaution.−

Government policies and schemes (like Pradhan Mantri Awas Yojana - PMAY) aim to provide affordable housing for all. These include financial assistance, slum redevelopment and public-private partnerships to bridge the housing gap in rural and urban areas.

Answer framework: List the major components of the PMAY housing scheme; describe the role of government in promoting cost-effective habitat technology.

Important point: Policy implementation involves legal and financial regulations. Do not treat a study-level summary as an authorised guide for housing finance or legal eligibility.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Human Settlements and Housing Needs.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Cost-Effective and Local Building Materials.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Laurie Baker's Innovative Techniques.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Climate-Responsive Design and Housing Policy.

8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Evolution of human settlements; urbanization and its impacts; housing as a basic need; assessment of housing shortage; principles of sustainable habitat development; UN Sustainable Development Goals (SDG 11)..
2. Develop a complete answer or practical plan for: Mud as a building material (stabilized soil blocks, cob, adobe); bamboo and its structural applications; laterite and local stone; industrial by-products (fly ash, GGBS); recycled aggregates and C&D waste..
3. Develop a complete answer or practical plan for: Laurie Baker's philosophy (simplicity, localism, economy); Rat-trap bond masonry; filler slabs (types of fillers); cost-effective arches, domes and vaults; brick jalis and ornamentation..
4. Develop a complete answer or practical plan for: Climatic zones and building design; thermal comfort; passive heating and cooling; energy efficiency in buildings; rural and urban housing policies; affordable housing schemes (PMAY)..

LAST-MILE REVISION

Quick Revision

Module 1: Human Settlements and Housing Needs

Evolution of human settlements; urbanization and its impacts; housing as a basic need; assessment of housing shortage; principles of sustainable habitat development; UN Sustainable Development Goals (SDG 11).

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Cost-Effective and Local Building Materials

Mud as a building material (stabilized soil blocks, cob, adobe); bamboo and its structural applications; laterite and local stone; industrial by-products (fly ash, GGBS); recycled aggregates and C&D waste.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Laurie Baker's Innovative Techniques

Laurie Baker's philosophy (simplicity, localism, economy); Rat-trap bond masonry; filler slabs (types of fillers); cost-effective arches, domes and vaults; brick jalis and ornamentation.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Climate-Responsive Design and Housing Policy

Climatic zones and building design; thermal comfort; passive heating and cooling; energy efficiency in buildings; rural and urban housing policies; affordable housing schemes (PMAY).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Settlements and urbanization	Human settlements evolve from rural to urban forms, influenced by economic, social and environmental factors.
Sustainable habitat principles	Sustainable habitats balance human needs with environmental protection.
Earth and bamboo construction	Earth (mud) is a traditional, low-embodied-energy material.
Local and alternative materials	Laterite and local stones reduce transport costs.
Rat-trap bond and filler slabs	Rat-trap bond is a brick-laying method that creates a cavity, saving bricks and mortar while providing thermal insulation.
Arches, domes and jalis	Arches and domes use the compressive strength of bricks to span openings and spaces cost-effectively.
Climate-responsive architecture	Design must respond to local climate (hot-dry, warm-humid, etc.
Housing policies and schemes	Government policies and schemes (like Pradhan Mantri Awas Yojana - PMAY) aim to provide affordable housing for all.

LEARNING RESOURCES

References

Recommended habitat-technology references

1. Laurie Baker, Cost-Effective Building Techniques.
2. Gautam Bhatia, Laurie Baker: Life, Work and Writings.
3. A. K. Lal, Handbook of Low Cost Housing.
4. Current National Building Code (NBC) and Housing Policy documents.

Approved public habitat-technology sources

- https://nptel.ac.in/courses/124107011
- https://onlinecourses.swayam2.ac.in/noc20_ar01/preview

These sources provide the verified study basis while official Course 6013D detail is unavailable; they do not replace official building codes, authorised structural designs, certified material specifications or authorised urban-planning data.